Employee Salary Calculator

Test Case Screenshots

PS No. 10 - B.Tech CSE | Python 3

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Test Case 1: Normal Input - PASS

Test Case 2: Boundary / Minimum Input - PASS

Test Case 3: Invalid Input - PASS

Test Case 4: Different Valid Scenario - PASS

Given Test Cases

From the problem statement

<table>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>Basic salary: 30000</td><td>HRA: 6000 | DA: 3000 | Gross: 39000
PF: 3600 | Tax: 1950 | Net Salary: 33450</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>Basic salary: 10000 (no tax slab)</td><td>HRA: 2000 | DA: 1000 | Gross: 13000
PF: 1200 | Tax: 0 | Net Salary: 11800</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>Basic salary: -5000, then abc</td><td>Invalid salary. Enter a positive number.</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>Basic salary: 50000</td><td>HRA: 10000 | DA: 5000 | Gross: 65000
PF: 6000 | Tax: 6500 | Net Salary: 52500</td></tr>
</table>
</td></tr>
<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Take employee name, ID and basic salary as input.
3. Calculate HRA, DA, PF and tax using operators and conditions.
4. Create functions such as calculate_gross() and print_slip().
5. Print a neat salary slip and allow more employees using a loop.</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
</table>

Test Case 1: Normal Input

Output of salary_calculator.py

Test Case 1: Normal Input

```
===== Employee Salary Calculator =====

Enter employee name: Amogh Agrawal
Enter employee ID: E101
Enter basic salary: 30000

========================================
            SALARY SLIP
========================================
Employee Name : Amogh Agrawal
Employee ID   : E101
----------------------------------------
Basic Salary  : 30000
HRA (20%)     : 6000
DA (10%)      : 3000
Gross Salary  : 39000
----------------------------------------
PF (12%)      : 3600
Tax           : 1950
----------------------------------------
NET SALARY    : 33450
========================================
HRA: 6000 | DA: 3000 | Gross: 39000
PF: 3600 | Tax: 1950 | Net Salary: 33450

Calculate for another employee? (y/n): n
Thank you!
```

Result: PASS - Basic 30000 -> Net Salary 33450

Test Case 2: Boundary / Minimum Input

Output of salary_calculator.py

Test Case 2: Boundary / Minimum Input

```
===== Employee Salary Calculator =====

Enter employee name: Amogh Agrawal
Enter employee ID: E102
Enter basic salary: 10000

========================================
            SALARY SLIP
========================================
Employee Name : Amogh Agrawal
Employee ID   : E102
----------------------------------------
Basic Salary  : 10000
HRA (20%)     : 2000
DA (10%)      : 1000
Gross Salary  : 13000
----------------------------------------
PF (12%)      : 1200
Tax           : 0
----------------------------------------
NET SALARY    : 11800
========================================
HRA: 2000 | DA: 1000 | Gross: 13000
PF: 1200 | Tax: 0 | Net Salary: 11800

Calculate for another employee? (y/n): n
Thank you!
```

Result: PASS - Basic 10000 (no tax slab) -> Net Salary 11800

Test Case 3: Invalid Input

Output of salary_calculator.py

Test Case 3: Invalid Input

```
===== Employee Salary Calculator =====

Enter employee name: Amogh Agrawal
Enter employee ID: E103
Enter basic salary: -5000
Invalid salary. Enter a positive number.
Enter basic salary: abc
Invalid salary. Enter a positive number.
Enter basic salary: 30000

========================================
            SALARY SLIP
========================================
Employee Name : Amogh Agrawal
Employee ID   : E103
----------------------------------------
Basic Salary  : 30000
HRA (20%)     : 6000
DA (10%)      : 3000
Gross Salary  : 39000
----------------------------------------
PF (12%)      : 3600
Tax           : 1950
----------------------------------------
NET SALARY    : 33450
========================================
HRA: 6000 | DA: 3000 | Gross: 39000
PF: 3600 | Tax: 1950 | Net Salary: 33450

Calculate for another employee? (y/n): n
Thank you!
```

Result: PASS - -5000 and abc rejected with: Invalid salary. Enter a positive number.

Test Case 4: Different Valid Scenario

Output of salary_calculator.py

Test Case 4: Different Valid Scenario

```
===== Employee Salary Calculator =====

Enter employee name: Amogh Agrawal
Enter employee ID: E104
Enter basic salary: 50000

========================================
            SALARY SLIP
========================================
Employee Name : Amogh Agrawal
Employee ID   : E104
----------------------------------------
Basic Salary  : 50000
HRA (20%)     : 10000
DA (10%)      : 5000
Gross Salary  : 65000
----------------------------------------
PF (12%)      : 6000
Tax           : 6500
----------------------------------------
NET SALARY    : 52500
========================================
HRA: 10000 | DA: 5000 | Gross: 65000
PF: 6000 | Tax: 6500 | Net Salary: 52500

Calculate for another employee? (y/n): n
Thank you!
```

Result: PASS - Basic 50000 -> Net Salary 52500